

Arihant Jain

Senior Backend Engineer · Distributed Systems & Platform Infrastructure · Python · AWS

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EXPERIENCE

Perfios (Karza Technologies)

Oct 2022 – Present

Senior Software Engineer

Apr 2024 – Present | New Delhi, IN (Remote)

- Delivered **\$700K+ in cumulative cost savings since Apr 2024** by rebuilding source routing and caching to cut third-party vendor dependency by **55%**, alongside retirement of redundant infrastructure.
- Owned **active-passive disaster recovery** for our KYC platform across Mumbai and Hyderabad, supporting **5M+ daily requests** at **99.9%+ measured availability**.
- Drove adoption of shared platform packages – secrets access, schema validation, and request configuration – across **200+ production APIs**, removing **300–400 lines of duplicated code** and turning changes that once required redeploying every API into a **single package update**.
- Built proxy-routing infrastructure for **120+ APIs** with health-aware selection and automatic cooldown recovery, **recovering 21% of requests** that would otherwise fail and eliminating **12–20 manual interventions per week**.
- Built a Redis-backed session library using **distributed locks** to coordinate authentication across concurrent Lambda workloads, **eliminating 35% of session-related failures** affecting **15K–30K daily requests**.
- Created maintenance-aware monitoring across daily, hourly, and event-driven checks that flags traffic, latency, and error anomalies against seven-day baselines and reprioritizes unhealthy dependencies automatically, **catching degradations before they escalate into incidents**.
- Reviewed **60+ pull requests** and established triage and debugging practices with junior and support engineers, **cutting incident triage SLA from 3 days to 3–4 hours**.
- Replaced a dedicated EC2 deployment VM with multi-region GitLab CI/CD pipelines with SAST for Lambda deployments.

Software Engineer I

Oct 2022 – Apr 2024 | Mumbai, IN (Remote)

- Re-architected a Lambda workflow as a Redis-backed Celery service on AWS Fargate; session reuse removed repeated logins and **tripled throughput**, while caching and rate limits enabled full retirement of an external vendor dependency.
- Optimized legacy KYC APIs, reducing **P99 latency from 8s to 3s** while improving reliability for client-facing verification workflows.
- Led migration of KYC and utility services from **Python 3.9 to 3.12**, containerizing **245 of 260 APIs** for cloud and on-premises deployment with **zero production rollbacks**.
- Built a shared vendor-billing and invoicing service adopted across **multiple products**, improving billing accuracy and operational visibility.
- Shipped **30+ APIs within six months** and maintained **100+ production APIs** spanning **300+ verification and data integrations**.

Software Engineer

Mar 2021 – Sep 2022

InTimeTec

Jaipur, IN (Onsite)

- Modernized a legacy .NET desktop application into an Angular web platform spanning **9 modules**, tripling operational throughput.
- Owned end-to-end delivery and client communication across a **4-engineer pod**, shortening deployment cycles by **44%** and increasing test coverage by **54%** through CI/CD and automated testing.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), SQL
Backend & APIs	FastAPI, Node.js, REST APIs, Pydantic
Data & Messaging	Redis, MongoDB, PostgreSQL, Celery, Kafka, RabbitMQ, Amazon SQS
Cloud & Delivery	AWS Lambda, Fargate, ECS, S3, EC2, CloudFormation, Docker, Linux, GitLab CI/CD, CloudWatch Logs Insights
Systems Architecture	Microservices, Event-Driven Architecture, Asynchronous Processing, Distributed Locking, Multi-Region DR
Testing & Security	Pytest, Playwright, Selenium, VAPT workflows

PROJECTS

Distributed Quota-Metering Engine

FastAPI · Redis Lua · MongoDB · Docker Compose

GitHub

- Engineered a per-organization, per-feature quota service where a **single Redis Lua script** handles validation, idempotency, deduction, and denial recording atomically, with all-or-nothing batch semantics, idempotent refunds, and cron-free monthly resets.
- Validated correctness with **10,000 simulated requests and zero over-serving**; retries using the same idempotency key consumed quota only once.

EDUCATION

Swami Keshvanand Institute of Technology

Aug 2017 – Jun 2021

B.Tech, Information Technology

Jaipur, IN

- Secured **All India Rank 91** in Round 1 of TCS CodeVita Season 9 (2020).